

Data Engineering Project

1. Title:End-to-End E-Commerce Data Platform for Sales and Customer Analytics

2. Group Members:

- **Vishwa Narayanaswamy:** Handles data collection, cleaning, SQL analysis, and dashboard creation.Designs warehouse tables and supports testing and documentation.
- **Koushik Raj Singh:** Designs the pipeline architecture and builds automated ETL workflows.Manages Airflow scheduling, Docker setup, deployment, and monitoring.

3. Problem Statement:

E-commerce businesses generate large amounts of data, but without a proper system it is hard to use effectively. This project builds an automated platform to organize data and support smarter decisions.

4. Introduction:

E-commerce has become a major part of everyday life, with people shopping online for convenience and variety. Every search, click, and purchase creates valuable data that can help businesses understand customer behavior, track sales, and improve services. When used properly, this data helps companies make faster and smarter decisions.

However, in many organizations, data is stored in different systems such as order databases, customer records, and product catalogs. Because the data is scattered, it becomes difficult to combine and analyze it quickly. Many businesses still rely on manual reports using spreadsheets, which takes time and can lead to errors. As the amount of data increases, this approach becomes slow and inefficient.

This project focuses on building an End-to-End E-Commerce Data Platform to solve these problems. The system automatically collects data from multiple sources, cleans and transforms it, and stores it in an organized way. The processed data is then presented through dashboards and reports that show insights like sales trends, customer behavior, and product performance. By automating the entire process, the platform ensures accurate and up-to-date information.

Overall, this project shows how data engineering can turn raw data into useful insights that help businesses improve performance and make better decisions.